



Hydrolysis of waste polyethylene terephthalate catalyzed by easily recyclable terephthalic acid

Weisheng Yang^a, Rui Liu^a, Chang Li^a, Yang Song^a, Chaoquan Hu^{a,b,*}

^a Nanjing IPE Institute of Green Manufacturing Industry, Nanjing, Jiangsu 211135, China

^b State Key Laboratory of Multiphase Complex Systems, Institute of Process Engineering, Chinese Academy of Sciences, Beijing 100190, China

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ABSTRACT

Hydrolysis of polyethylene terephthalate (PET) is an efficient strategy for the depolymerization of waste PET to terephthalic acid (TPA), which can be used as a fundamental building block for the repolymerization of PET or for the synthesis of biodegradable plastics and metal–organic frameworks. However, most of the reported hydrolysis catalysts are strong acids or bases, which are soluble in reaction media and difficult to separate after the reaction, leading to high production costs and a profound influence on the environment. Herein, we propose the use of TPA, the basic unit of PET, as an acid catalyst to promote the hydrolysis of PET. Under optimized conditions, i.e., 2.5 g of PET, a TPA concentration of 0.1 g/mL, mass ratio PET:H₂O of 1:8, 220 °C of temperature, and 180 min of reaction time, a PET conversion of up to 100.0% and a TPA yield of 95.5% were achieved. Furthermore, the produced TPA exhibited a high purity of 99%, similar to that of fresh TPA, and was easily recoverable for PET hydrolysis without tedious workup and purification processes. More importantly, the hydrolysis efficiency was maintained over eight consecutive reaction cycles. Overall, this study provides a green, easy, and low-cost technology to recover and reuse TPA for waste PET hydrolysis.

1. Introduction

As an essential engineering plastic, polyethylene terephthalate (PET) is widely used in polyester fibers, food packaging, plastic films, electronic devices, mechanical equipment, and other fields because of its high transmittance, excellent chemical stability, barrier properties, and mechanical properties (Karayannidis et al., 2006; Ke et al., 1999). In recent years, with the continuous economic growth, global PET consumption has gradually increased every year (Furukawa et al., 2018). The huge demand for PET plastic products, along with their short life cycle, generates a large amount of discarded PET plastics, which severely damages the ecological environment and results in a waste of resources if not appropriately processed (Gibb, 2019; Sánchez & Colinson, 2011). In this regard, the recycling of waste PET plastics can be envisaged as an effective approach to extend their life cycle, reduce the consumption of nonrenewable resources, prevent plastic pollution, and protect the ecological environment (Kenny et al., 2008; Raheem et al., 2019).

Mechanical recycling is currently the most common method for waste PET recycling, which includes grinding and melt-processing of PET waste to obtain recycled PET particles (Ragaert et al., 2017;

Thiounn and Smith, 2020). However, this process is still limited by the need for conducting classification and pretreatment of waste plastics and by the loss of quality in the recycled product caused by the degradation of plastics during mechanical treatment (López et al., 2014). Compared with physical methods, biochemical/enzymatic recycling (Tournier et al., 2020) and chemical recycling methods can realize the permanent recycling of waste PET, which is more in line with the current demand for a sustainable development and has a number of advantages and development prospects (Coates & Getzler, 2020; Hong & Chen, 2017; Schneiderman & Hillmyer, 2017; Vollmer et al., 2020). Hydrolysis (Wang et al., 2019; Yoshioka et al., 1994), methanolysis (Du et al., 2020; Pham & Cho, 2021), glycolysis (Liu et al., 2018; Liu et al., 2020; Zhou et al., 2019), and other processes (Mittal et al., 2010; Yoshida et al., 2016) have been proposed as chemical recycling methods. Hydrolysis, the most facile method for PET treatment, comprises PET depolymerization to terephthalic acid (TPA) and ethylene glycol (EG). These monomers are the basic raw materials for the repolymerization of PET. In general, the hydrolysis of PET is promoted by acid and base catalysts such as sulfuric acid (Campanelli et al., 1993), nitric acid (Yoshioka et al., 1998), and sodium hydroxide (Ügdüler et al., 2020). Unfortunately, large amounts of strongly acidic/alkali wastewater are produced

* Corresponding author at: Nanjing IPE Institute of Green Manufacturing Industry, Nanjing, Jiangsu 211135, China.

E-mail address: cqhu@ipe.ac.cn (C. Hu).

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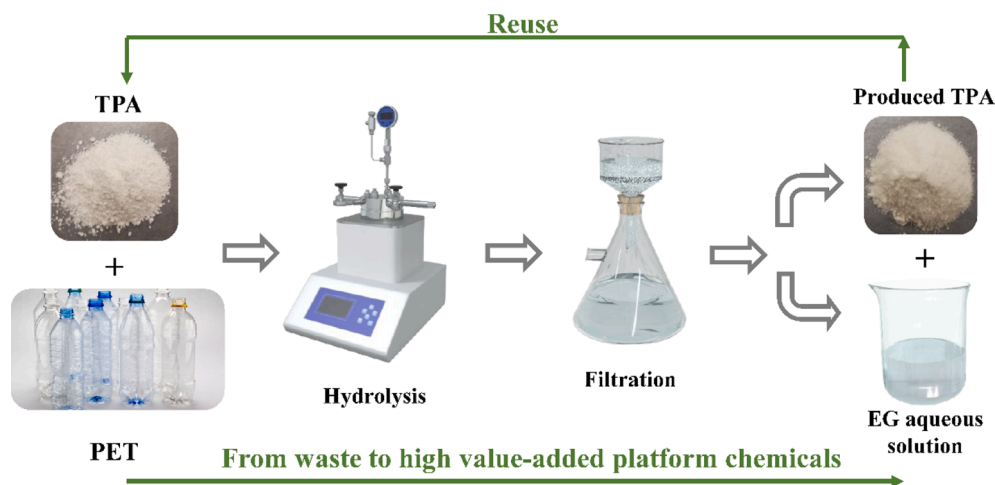


Fig. 1. Schematic diagram of the experimental process for the hydrolysis of polyethylene terephthalate (PET) catalyzed by easily recyclable terephthalic acid (TPA).

in these reaction systems, causing tremendous pressure on the environment with a similar impact to that caused by the accumulation of PET polyester bottles (Kang et al., 2020). In addition, the acid and alkali catalysts used in the hydrolysis reaction are difficult to recover and reuse, resulting in a waste of resources and an increase in the production costs (Kang et al., 2020).

The development of a simply recoverable catalyst for PET hydrolysis is bound to significantly reduce costs, ease the pressure on the environment during the hydrolysis process, and promote its industrialization process. However, reports on recyclable catalysts for the hydrolysis of PET are still scarce. Cao et al. reported a Tungsten-promoted titania solid acid catalyst for PET hydrolysis in supercritical CO₂. At 160 °C and 15 MPa for 15 h, PET was completely degraded to TPA, EG, and DEG, and the catalyst could be recycled and reused five times without noticeable performance loss (Guo et al., 2016). Cha and Kim et al. utilized ZSM-5-based zeolites as easily recoverable acidic catalysts for PET hydrolysis (Kang et al., 2020). After six cycles, the ZSM-5-based zeolites were easily recycled as a fresh catalyst without loss of catalytic activity. However, these solid catalysts were difficult to separate from the PET hydrolysis products (TPA) only by simple filtration due to their poor aqueous solubility, requiring the dissolution of TPA using a sodium hydroxide solution and the addition of a strong acid to regenerate TPA after filtering the solid catalysts. As a result, the overall process generates additional costs and a great deal of salty wastewater.

This paper describes the use of TPA ($pK_{a1} = 3.54$, $pK_{a2} = 4.46$), the main PET hydrolysis product, as a solid organic acidic catalyst for waste PET hydrolysis, thereby avoiding unnecessary separation processes and the production of large amounts of salty wastewater (Fig. 1). The effect of the reaction conditions (temperature, time, TPA concentration, and PET:H₂O mass ratio) and the kinetics of PET hydrolysis were investigated. Besides, the catalyst recoverability was studied by performing eight cycles of recycling and reuse. This work provides a simple, efficient, green, and sustainable new technology for recycling waste PET plastics.

2. Materials and methods

2.1. Materials

Waste PET granules of 60 mesh were obtained by milling colorless PET bottles. The intrinsic viscosity of the PET sample was determined to be 0.7 dL/g using an Ubbelohde viscometer. The crystallinity of waste PET was calculated to be 32.5% through Differential Scanning Calorimetry (DSC). TPA was purchased from Aladdin Biochemical Co., Ltd (Shanghai, China). Sulfuric acid and sodium hydroxide were provided by Nanjing Chemical Reagent Co. Ltd (Nanjing, China). Deionized water

was used in this study.

2.2. General hydrolysis of PET

In a typical experiment, 2.5 g of PET, 20.0 g of water, and 2.0 g of TPA were added into a 100 mL pressure-tight reactor with a magnetic stirrer. The hydrolysis of PET was conducted at a reaction temperature ranging from 180 °C to 220 °C and a reaction time from 60 to 240 min. After the reaction, the mixture solution was filtrated, affording a filter cake consisting of undepolymerized PET and TPA and a filtrate containing water and EG. The collected filter cake was washed by 1 mol/L sodium hydroxide solution and filtrated. The produced TPA was dissolved in the form of sodium-TPA, and undepolymerized PET remained as a residue. The undepolymerized PET was then collected, dried, and weighed. The PET conversion was calculated according to Equation (1) as follows:

$$\text{PET conversion} = \frac{W_0 - W_2}{W_0} \times 100\% \quad (1)$$

where W_0 is the initial weight of PET, and W_2 is the weight of undepolymerized PET.

Then, sulfuric acid was added into the dissolved sodium-TPA aqueous solution to generate pure TPA. The obtained TPA was dried at 60 °C for 12 h and then weighed. The TPA yield was calculated according to Equation (2),

$$\text{TPA yield} = \frac{W_3 - W_1}{W_4} \times 100\% \quad (2)$$

where W_1 , W_3 , and W_4 are the weight of TPA added as the catalyst, the weight of produced pure TPA, and the theoretical weight of TPA, respectively.

2.3. Cyclic stability test

2.5 g of PET, 20.0 g of water, and 2.0 g of TPA were added to a pressure-tight reactor with a magnetic stirrer. The reaction was performed under 220 °C for 180 min. Under this condition, waste PET was completely degraded, and only solid TPA and EG aqueous solutions were generated. Consequently, high-purity TPA could be separated by simple filtration, thereby avoiding the processes of alkali dissolution and acid precipitation. The solid product was collected and dried at 60 °C for 12 h and named as TPA-cycle-0. This process was repeated with 2.0 g of TPA-cycle-0 instead of fresh TPA. The product was named as TPA-cycle-1. After eight repetitions, TPA-cycle-8 was finally obtained. For the cyclic stability test, the TPA yield was calculated according to Equation (3) as follows:

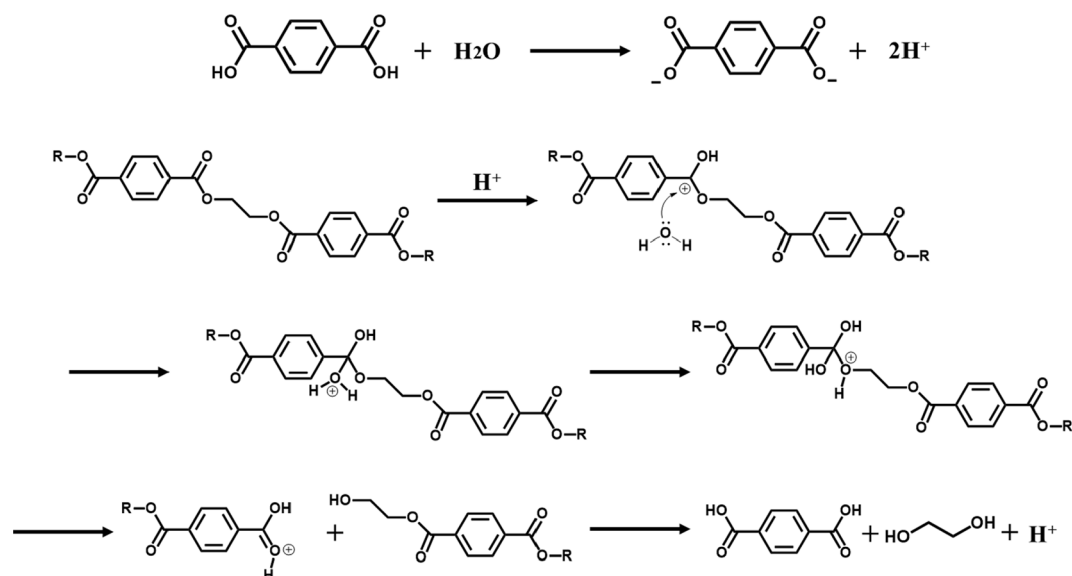


Fig. 2. PET hydrolysis mechanism with TPA as an acid catalyst.

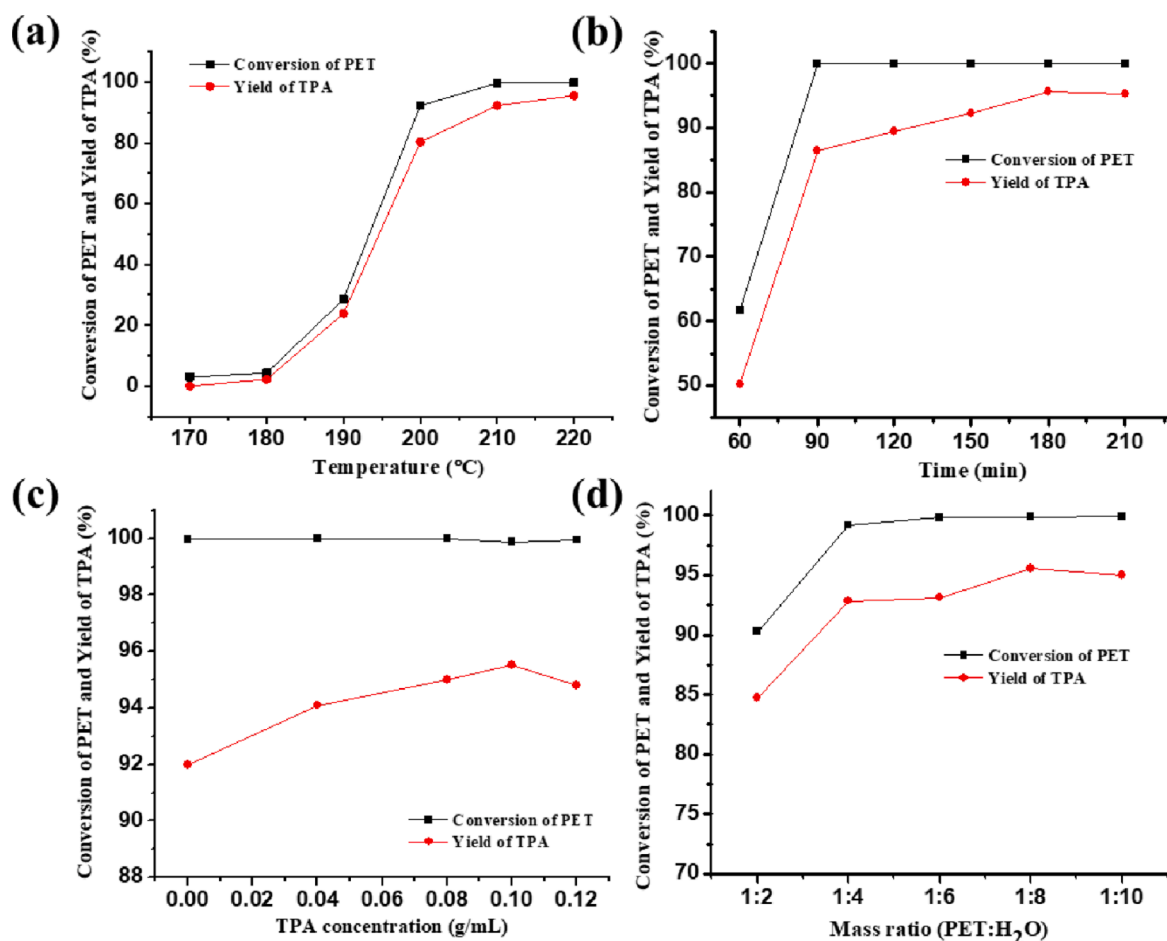


Fig. 3. Optimization of polyethylene terephthalate (PET) hydrolysis reaction conditions. (a) Influence of the reaction temperature. Reaction conditions: 2.5 g of PET, terephthalic acid (TPA) concentration of 0.1 g/mL, PET:H₂O mass ratio of 1:8, 180 min. (b) Influence of the reaction time. Reaction conditions: 2.5 g of PET, TPA concentration of 0.1 g/mL, PET:H₂O mass ratio of 1:8, 220 °C. (c) Influence of the TPA concentration. Reaction conditions: 2.5 g of PET, PET:H₂O mass ratio of 1:8, 220 °C, 180 min. (d) Influence of the PET:H₂O mass ratio. Reaction conditions: 2.5 g of PET, TPA concentration of 0.1 g/mL, 220 °C, 180 min.

$$\text{TPA yield} = \frac{W_3 P - W_1}{W_4} \times 100\%$$

(3)

where W_1 , W_3 , W_4 , and P are the weight of TPA added as the catalyst, the weight of produced TPA, the theoretical weight of TPA, and the purity of produced TPA, respectively.

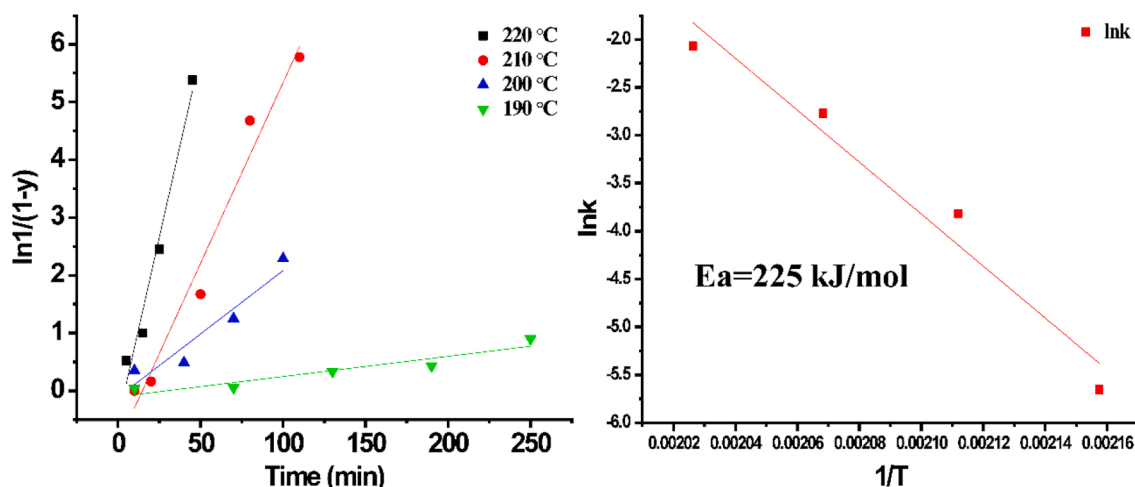


Fig. 4. (a) Rate of polyethylene terephthalate (PET) hydrolysis at different temperatures. (b) Arrhenius plot of the rate constant of PET hydrolysis (reaction conditions: 2.5 g of PET, terephthalic acid (TPA) concentration of 0.1 g/mL, PET:H₂O mass ratio of 1:8).

2.4. Characterization

The crystal structures of PET and produced TPA were investigated by powder X-ray diffraction (XRD; Smartlab 9kw, Rigaku, Japan), using Cu K α radiation. The diffraction patterns were collected with a 2θ range between 5° and 60° at a scan speed of 5°/min. Fourier transform infrared (FTIR) spectroscopy measurements of PET were performed on a VERTEX 80 V FTIR spectrometer (Bruker, Germany) equipped with a universal attenuated total reflection probe. ¹H and ¹³C nuclear magnetic resonance (NMR) spectra of produced TPA were recorded on a Bruker 400 MHz NMR spectrometer in *d*₆-dimethylsulfoxide solution. The purity of produced TPA was analyzed by ultraviolet–visible spectrophotometry (UV–vis). First, 0.1 g of TPA purchased from Aladin was weighed and dissolved in 6 mL of a 1 mol/L sodium hydroxide solution. The sodium terephthalate (NaTPA) solution was then diluted to a series of certain concentrations to produce standard solutions of TPA (0.000 mg/mL, 0.004 mg/mL, 0.008 mg/mL, ..., 0.02 mg/mL). The standard curve was drawn by recording the absorbance of the standard solutions at 240 nm. To detect the purity of produced TPA, a solution of NaTPA was prepared by the same procedure as the standard solutions.

3. Results and discussion

3.1. PET hydrolysis catalyzed by TPA

When using TPA as an acidic catalyst, the carboxyl groups are ionized into hydrogen ions in an aqueous solution to catalyze PET hydrolysis. As illustrated in Fig. 2, the hydrogen ions then react with the carbonyl groups of PET, generating hydroxyl groups and carbocations (Kumagai et al., 2015; Li et al., 2015; Sinha et al., 2010). These carbocations are susceptible to nucleophilic attack by water, reducing the thermodynamic activation energy (E_a) and increasing the reaction kinetics (Sinha et al., 2010). Following the cleavage of the esters of PET, TPA and EG are generated as PET hydrolysis products (Sammon et al., 2000). In the present study, various parameters affecting the hydrolysis of PET were evaluated, and the results are shown in Fig. 3. Fig. 3a presents the influence of the reaction temperature on the conversion of PET and the yield of TPA. The results indicate that the reaction temperature plays a critical role in PET hydrolysis. Thus, when the reaction temperature was as low as 170 °C–180 °C, PET was hard to degrade. Upon increasing the temperature, the conversion of PET and the yield of TPA improved significantly because the reaction rate of PET hydrolysis also increased. The PET conversion and the TPA yield reached maximum values of 100.0% and 95.5%, respectively, at 220 °C. Fig. 3b displays the influence of the reaction time on the reaction conversion and yield.

When the reaction time was 90 min, the PET conversion reached 100.0%, but the TPA yield was only 86.4%. The yield increased slowly as the reaction proceeded, reaching a maximum value of 95.5% at 180 min of reaction time and remaining constant due to the chemical equilibrium between the TPA monomer and the oligomer. To study the effect of the catalyst concentration on the PET hydrolysis, the reaction was performed with a TPA concentration of 0–0.12 g/mL, and the results are shown in Fig. 3c. As the acid concentration increased, the yield of TPA increased continuously. When the acid catalyst concentration reached 0.1 g/mL, PET was depolymerized entirely, and the TPA yield reached the maximum value. As shown in Fig. 2, the presence of water molecules could promote PET hydrolysis. The influence of the PET:H₂O mass ratio is shown in Fig. 3d. The TPA yield was found to increase with increasing the water content, and the optimal results were obtained for a PET:H₂O mass ratio of 1:8. After a comprehensive study of the factors affecting PET hydrolysis, the optimized reaction conditions were set as follows: 2.5 g of PET, a TPA concentration of 0.1 g/mL, a PET:H₂O mass ratio of 1:8, 220 °C of temperature, and 180 min of reaction time.

3.2. Kinetic of PET hydrolysis catalyzed by TPA

PET hydrolysis catalyzed by an acid has been previously reported to follow first-order kinetics (Kang et al., 2020; Li et al., 2015). The kinetic equation of PET hydrolysis can be expressed by Equation (4):

$$\ln \frac{1}{1-y} = kt \quad (4)$$

where y , k , and t are the conversion of PET, the rate constant of the reaction, and the reaction time, respectively.

According to the rate constant of the reaction under various temperatures (190 °C, 200 °C, 210 °C, 220 °C), E_a can be obtained by the Arrhenius equation:

$$\ln k = \ln A - \frac{E_a}{RT} \quad (5)$$

where k , A , R , and T are the rate constant of the reaction, the pre-exponential factor, the gas constant (8.31 J/k mol), and the absolute temperature in Kelvin, respectively.

The effect of the reaction temperature on k calculated by Equation (4) is shown in Fig. 4a. The experimental data presented good linearity under various temperatures. In addition, upon increasing the temperature, k increased significantly, indicating that the temperature is a crucial influencing factor for PET hydrolysis. Fig. 4b presents an Arrhenius plot of the apparent k with a linear correlation factor ($R^2 = 0.96$). The apparent E_a of this reaction was determined to be 225 kJ/mol

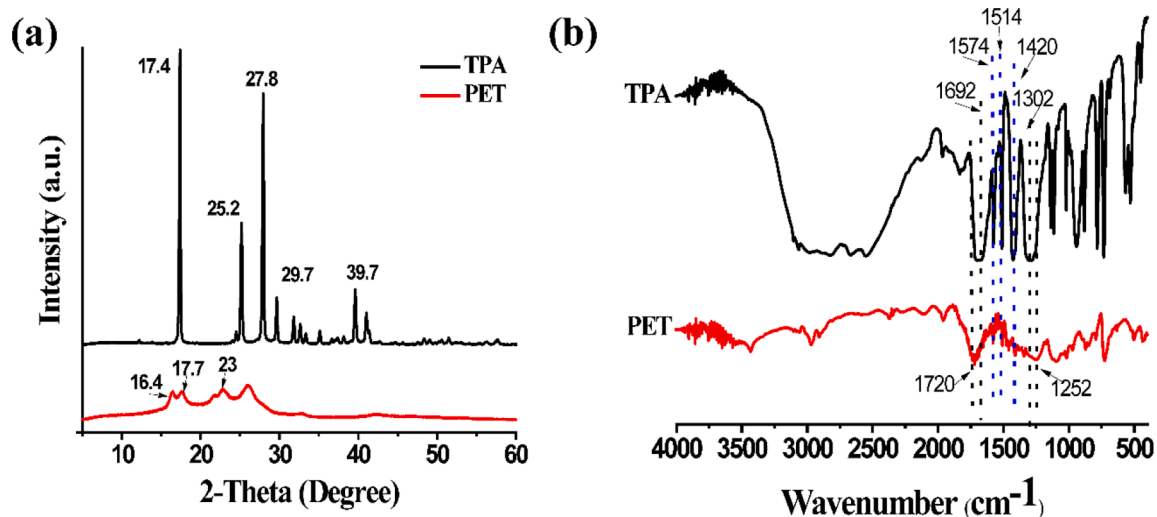


Fig. 5. (a) X-ray diffraction patterns and (b) Fourier transform infrared spectra of produced terephthalic acid (TPA) and waste polyethylene terephthalate (PET).

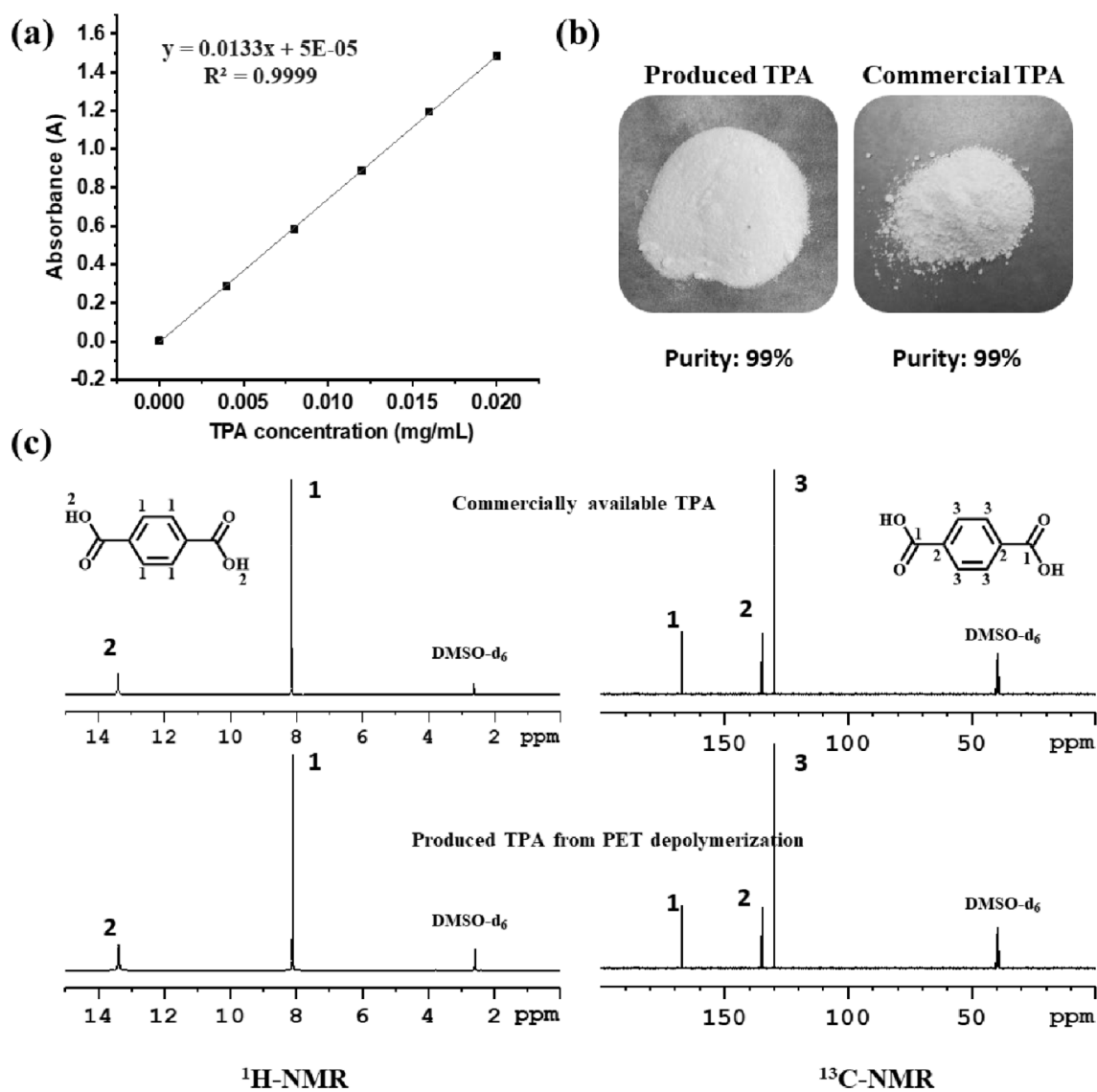


Fig. 6. (a) Terephthalic acid (TPA) ultraviolet absorption standard curve at a wavelength of 240 nm. (b) Digital photographs of the TPA produced from polyethylene terephthalate (PET) hydrolysis and commercial TPA. (c) ¹H and ¹³C nuclear magnetic resonance (NMR) spectra of the TPA produced from PET hydrolysis and commercial TPA.

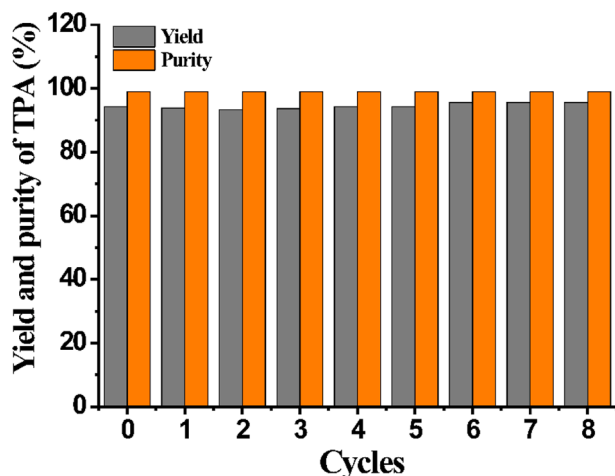


Fig. 7. Recycling and reuse of terephthalic acid (TPA) for polyethylene terephthalate (PET) hydrolysis (reaction conditions: 2.5 g of PET, TPA concentration of 0.1 g/mL, PET:H₂O mass ratio of 1:8, 220 °C, 180 min).

from the slope of the Arrhenius plot. Low E_a favor larger rate constants, and thus speed up the reaction. Through the reaction kinetics study, it can be found that the hydrolysis of PET catalyzed by TPA has a higher E_a of 225 kJ/mol than sulfuric acid (89 kJ/mol) (Yoshioka et al., 2001), nitric acid (101 kJ/mol) (Yoshioka et al., 1998), and hydrochloric acid (75–113 kJ/mol) (Ravens, 1960), which is mainly due to its relatively

weak acidity. Therefore, using TPA as the acid catalyst requires more severe conditions to achieve effective hydrolysis of waste PET. Nevertheless, compared with strong acid, utilization of TPA as catalyst for PET hydrolysis still has many advantages, such as recyclability, easily achieving product separation, and wastewater free.

3.3. Characterization and analysis of PET and degradation products

The XRD diffraction spectra of produced TPA and waste PET are shown in Fig. 5a. The spectrum of raw PET exhibited broad diffraction peaks at 2θ values of approximately 16.4°, 17.7°, and 23.0°, conforming to a typical diffraction pattern of a polyester crystal structure (Liu et al., 2020). Meanwhile, the spectrum of the TPA monomer showed some sharp and strong diffraction peaks at $2\theta = 17.4^\circ, 25.2^\circ, 27.8^\circ, 29.7^\circ,$ and 39.7° (Kaur et al., 2020), indicating a high crystallinity for the TPA product of PET hydrolysis. These peaks were consistent with those observed for commercial TPA and previous literature (Dyosiba et al., 2019). FTIR was conducted to analyze the chemical structure of produced TPA and raw PET, and the details were displayed in Fig. 5b. The peaks at 1574, 1514, and 1420 cm^{-1} can be attributed to the vibration of the benzene ring of PET and TPA (Wang et al., 2019). For raw PET, the C = O stretching vibration and C–O vibration of the ester bonds were observed at 1720 and 1252 cm^{-1} , respectively (Liu et al., 2020). Moreover, the peaks at 1692 and 1302 cm^{-1} can be assigned to the COOH group of TPA (Wang et al., 2019).

It can be seen from the standard absorption curve of TPA depicted in Fig. 6a that the TPA concentration had a good linear correlation with the

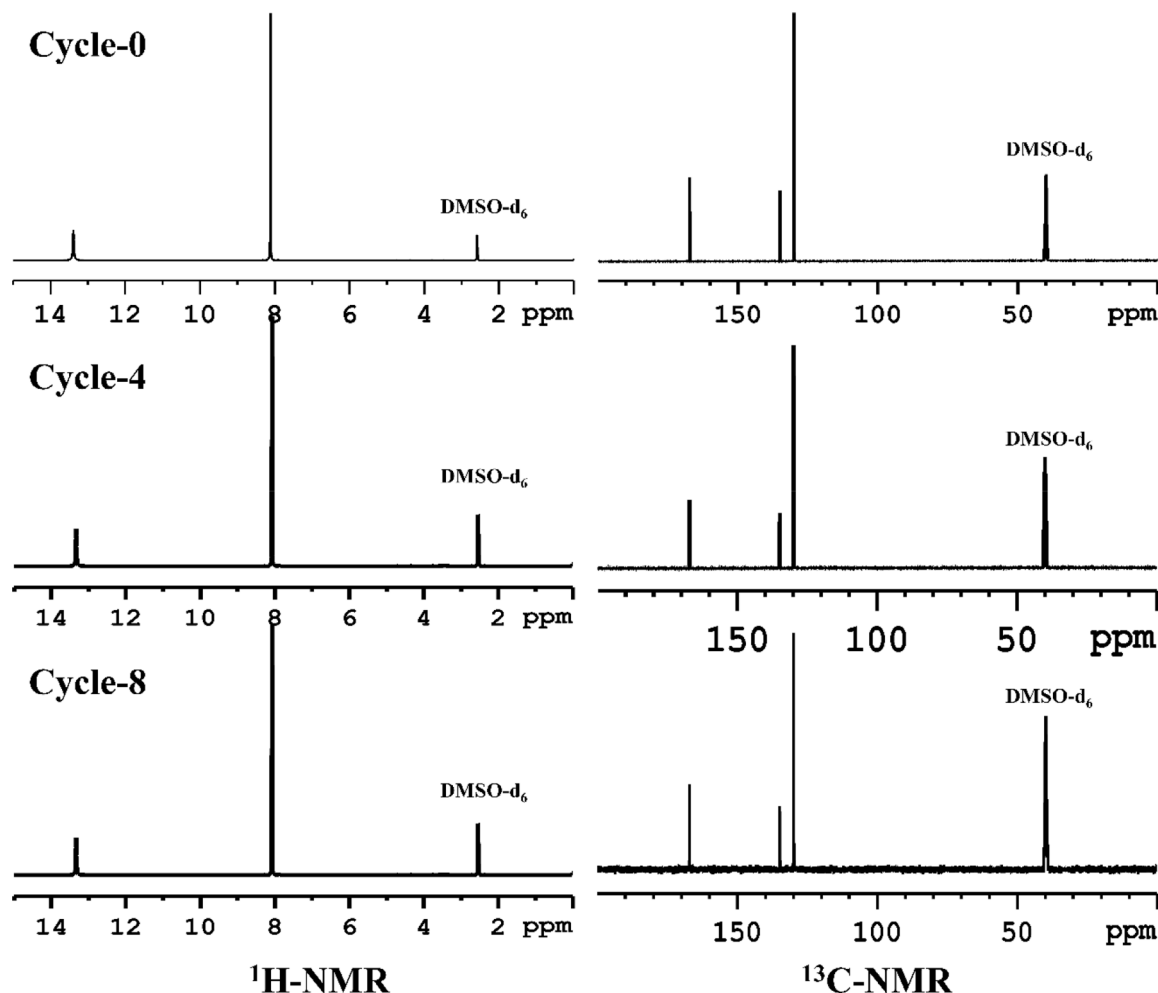


Fig. 8. ¹H and ¹³C nuclear magnetic resonance (NMR) spectra of produced terephthalic acid obtained at different cycles.

absorbance ($R^2 = 0.9999$) at a wavelength of 240 nm, with an equation of $y = 0.0133x + 5E - 05$. According to the TPA standard curve, the quality and purity (99%) of the TPA produced from PET hydrolysis were determined to be comparable to those of commercial TPA (Fig. 6b). In addition, the ^1H and ^{13}C NMR spectra of produced TPA and commercially available TPA were recorded to analyze the chemical structure and purity of the former (Fig. 6c). In the ^1H NMR spectra, the signals at 8.1 and 13.4 ppm are ascribable to the aromatic protons of the benzene ring and the COOH proton, respectively (Strukil, 2021). Similarly, in the ^{13}C NMR spectra, three peaks appearing at 130.1, 135.0, and 167.1 ppm can be assigned to the aromatic (denoted as C3), quaternary aromatic (denoted as C2), and carbonyl (denoted as C1) carbon atoms of TPA (Pétiaud et al., 1992). These results indicate that the TPA produced from PET hydrolysis had a consistent structure with that of commercially available TPA. Besides, no other peaks were observed in the NMR spectra, which is indicative of the high purity of produced TPA.

3.4. Recycling and reuse of TPA

From environmental and economic viewpoints, it is particularly important to develop recyclable and reusable catalysts for the hydrolysis of PET (Al-Sabagh et al., 2016). Common strong acid and alkali catalysts (e.g., sulfuric acid, nitric acid, and sodium hydroxide) for PET hydrolysis are difficult to recover and reuse due to their high solubility in water. In contrast, the low solubility of TPA in water enabled its easy recovery from the reaction solutions by filtration, thereby avoiding the generation of large quantities of wastewater. In this study, the produced TPA will be used as acid catalyst for the next hydrolysis reaction. The yield and purity of produced TPA were used to assess the cyclic stability of TPA, and the results are shown in Fig. 7. The produced TPA yield was maintained at 93.3%–95.6% over eight cycles of continuous reuse. In addition, produced TPA had a high purity of 99%, revealing that its repeated use had little effect on the PET hydrolysis efficiency. The chemical structure of the TPA obtained under different cycles was also investigated by ^1H and ^{13}C NMR spectroscopy. The results depicted in Fig. 8 indicate that the spectra of the TPA produced from different cycles and fresh TPA were consistent. After the eight cycles, the chemical structure and purity of the produced TPA remained unchanged and still showed the acid hydrolysis ability for PET degradation.

4. Conclusion

In summary, TPA can behave as an efficient catalyst for the PET hydrolysis process and can be easily recovered and reused. Under the optimized conditions, i.e., 2.5 g of PET, a TPA concentration of 0.1 g/mL, a PET:H₂O mass ratio of 1:8, 220 °C of reaction temperature, and 180 min of reaction time, PET was degraded entirely, and the produced TPA yield reached 95.5%. The kinetic reaction studies showed that the PET hydrolysis was a first-order reaction with an apparent E_a of 225 kJ/mol. Produced TPA had a high purity of 99%, almost the same structure as fresh TPA, and could be reused up to eight times without loss of the catalytic activity. Compared with the reported hydrolysis process using concentrated acid, this technology requires higher reaction temperature and longer reaction time to achieve same hydrolysis efficiency. However, the recyclability of TPA catalyst, easily achieving product separation and wastewater free make this technology shows a certain industrialization prospect. In the future research works, we will conduct a more in-depth study to promote its industrialization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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